Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EMT513S - MATLAB

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Write the command for area of circle?
2. Enter the matrix $A = \begin{bmatrix} 2 & 6 \\ 3 & 9 \end{bmatrix}$.
3. Give a Logical operator.
4. What is command for fitting polynomial of order n?
5. Write syntax for command 'quad'.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Write about Display formats.
7. A= [1 2 3; 4 5 6; 7 8 9] write the commands for A^T, A^2.
8. Write briefly on for loop command.
9. Plot the raw data using Plot command for the following data
   X= 0 0.15 0.2 0.3 0.4
   Y=0 1.6 2.2 3.4 4.8

10. Compute the integral $I = \int_{-1}^{1}\int_{0}^{2} 1 - 6x^2 y dx dy$.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Write about creating and working with Array of numbers.

(or)

b) Write about arrays.

12. a) Discuss about build in math functions.

(or)

b) Write a note on element by element operation.

13. a) Describe with example if-else-else statement.

(or)

b) Describe with example statement switch-case-otherwise.

14. a) Write about interpolation with example.

(or)

b) Discuss about plot the straight line using data.

15. a) Describe with example about ode45 function.

(or)

b) Solve the first order linear differential equation dx/dt = x+t, x(0)=0.

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